

Question 10.1 — pwd and exit

Question:

To retrieve the current directory, display the path, process exit requests, clean up resources, save state, and verify correct behavior.

Question 10.2 — export

Question:

To parse export syntax, update environment variables, validate names, handle existing variables, export to child processes, and test functionality.

```
saikrishna__90156@DESKTOP-2JF9H5T:~$ nano skill10.c
saikrishna__90156@DESKTOP-2JF9H5T:~$ gcc skill10.c -o skill10
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./skill10
shell> ls
Command not found.
shell> help
Command not found.
shell> pwd
/home/saikrishna__90156
shell> hello
Command not found.
shell> ^C
saikrishna__90156@DESKTOP-2JF9H5T:~$ nano skill10a.c
saikrishna__90156@DESKTOP-2JF9H5T:~$ gcc skill10a.c -o skill10a
saikrishna__90156@DESKTOP-2JF9H5T:~$ ./skill10a
shell> ls
Unknown command.
shell> hello
Unknown command.
shell> pwd
Unknown command.
shell> ^C
saikrishna__90156@DESKTOP-2JF9H5T:~$ nano skill10a.c
saikrishna__90156@DESKTOP-2JF9H5T:~$ gcc skill10a.c -o skill10a
skill10a.c:81:2: error: stray '#' in program
   81 | }#include <stdio.h>
      |           ^
skill10a.c:81:11: error: expected '=', ',', ';', 'asm' or '__attribute__' before '<' token
   81 | }#include <stdio.h>
      |           ^
skill10a.c:107:5: error: redefinition of 'main'
  107 | int main()
```

```
saikrishna__90156@DESKTOP-2JF9H5T: ~$ nano skill10a.c
skill10a.c:107:5: error: redefinition of 'main'
107 | int main()
    | ^~~~~
skill10a.c:27:5: note: previous definition of 'main' with type 'int(void)'
27 | int main()
    | ^~~~~
saikrishna__90156@DESKTOP-2JF9H5T: ~$ nano skill10b.c
saikrishna__90156@DESKTOP-2JF9H5T: ~$ gcc skill10b.c -o skill10b
saikrishna__90156@DESKTOP-2JF9H5T: ~$ ./skill10b
shell> pwd
Unknown command.
shell> hello
Unknown command.
shell> help
Unknown command.
shell> pwd
Unknown command.
shell> exit
Exiting shell...
saikrishna__90156@DESKTOP-2JF9H5T: ~$ cat skill10b.c
#include <stdio.h>
#include <stdlib.h>
#include <string.h>
#include <ctype.h>
#include <unistd.h>
#include <sys/wait.h>

#define SIZE 200

/* Check whether the variable name is valid */
int isValidName(const char *name)
{
    int i;
```



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```
/* First character must be alphabetic or underscore */
if (!isalpha((unsigned char)name[0]) && name[0] != '_')
    return 0;

/* Remaining characters can be letters, digits or underscore */
for (i = 1; name[i] != '\0'; i++)
{
    if (!isalnum((unsigned char)name[i]) && name[i] != '_')
        return 0;
}

return 1;
}

int main()
{
    char command[SIZE];

    while (1)
    {
        printf("shell> ");
        fflush(stdout);

        if (fgets(command, sizeof(command), stdin) == NULL)
            break;

        command[strcspn(command, "\n")] = '\0';

        /* exit */
        if (strcmp(command, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }
    }
}
```

```
}

/* export command */
if (strncmp(command, "export ", 7) == 0)
{
    char *assignment = command + 7;
    char *equalSign = strchr(assignment, '=');

    if (equalSign == NULL)
    {
        printf("Invalid syntax. Use: export NAME=VALUE\n");
        continue;
    }

    /* Separate NAME and VALUE */
    *equalSign = '\0';

    char *name = assignment;
    char *value = equalSign + 1;

    /* Validate variable name */
    if (!isValidName(name))
    {
        printf("Invalid variable name: %s\n", name);
        continue;
    }

    /* Set environment variable */
    if (setenv(name, value, 1) == -1)
    {
        perror("setenv");
        continue;
    }

    printf("Exported: %s=%s\n", name, value);
}
```

```

        printf("Exported: %s=%s\n", name, value);

        /* Demonstrate child-process inheritance */
        pid_t pid = fork();

        if (pid == -1)
        {
            perror("fork");
        }
        else if (pid == 0)
        {
            printf("Child process sees %s=%s\n",
                   name, getenv(name));
            exit(0);
        }
        else
        {
            waitpid(pid, NULL, 0);
        }
    }
    else
    {
        printf("Unknown command.\n");
    }
}

return 0;
}
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat skill10.c
#include <stdio.h>
#include <unistd.h>
#include <string.h>

```

```
saikrishna__90156@DESKTOP-2JF9H5T:~$ cat skill10.c
#include <stdio.h>
#include <unistd.h>
#include <string.h>

int main()
{
    char command[100];
    char path[1024];

    while (1)
    {
        printf("shell> ");

        if (fgets(command, sizeof(command), stdin) == NULL)
            break;

        command[strcspn(command, "\n")] = '\0';

        if (strcmp(command, "pwd") == 0)
        {
            if (getcwd(path, sizeof(path)) != NULL)
                printf("%s\n", path);
        }
        else if (strcmp(command, "exit") == 0)
        {
            printf("Exiting shell...\n");
            break;
        }
        else
        {
            printf("Command not found.\n");
        }
    }
}
```